

JEE MAIN REVISION MASTER: ISOLATION OF ELEMENTS OUT OF SYLLABUS

Subject: Chemistry | Target Scope: JEE Main Previous Years | Empirical Base: 106 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
Very High	Isolation Of Elements — Core Principles & Governing Laws	75 PYQs	Core Principle
Very High	Isolation Of Elements — Applied Cases, Reactions & Traps	28 PYQs	Applied Topics
Moderate	Isolation Of Elements — Standard Calculations & Problem Archetypes	4 PYQs	Quantitative Analysis

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
Isolation Of Elements — Core Principles & Governing Laws	$Al_2O_3, \quad Cu(OH)_2, CuCO_3, \quad \Delta_r S^0$
Isolation Of Elements — Applied Cases, Reactions & Traps	$CaCO_3 \xrightarrow{\Delta} CaO + CO_2, \quad CS_2, \quad Zr + 2I_2 \rightarrow ZrI_4$
Isolation Of Elements — Standard Calculations & Problem Archetypes	$^{\circ}, \quad Al_2O_3, \quad 900^{\circ}C$

3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Isolation Of Elements — Core Principles & Governing Laws (Very High Yield — 75 PYQs)

Core Principle: Fundamental definitions, core postulates, and governing physical/chemical relations in Isolation Of Elements.

■ **Caution / JEE Pitfall:** Carefully verify units, standard reference states, and sign conventions in Isolation Of Elements.

■ **Speed Trick:** Focus on high-yield formulas and standard limiting approximations tested in Isolation Of Elements PYQs.

Representative Source PYQs:

- [2025 Evening Shift] Q#1 (MCQ): "The purification method based on the following physical transformation is : $\text{X(s)} \xrightarrow{\text{Heat}} \text{Y(l)}$ " [Ans: N/A]
- [2023 Morning Shift] Q#1 (MCQ): "Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R Assertion A : In the Ellingham diagram,..." [Ans: N/A]
- [2023 Morning Shift] Q#1 (MCQ): "Which of the following is used as a stabilizer during the concentration of sulphide ores?..." [Ans: N/A]

3.2 Isolation Of Elements — Applied Cases, Reactions & Traps (Very High Yield — 28 PYQs)

Core Principle: Special cases, experimental observations, and conceptual traps designed to test deep understanding of Isolation Of Elements.

■ **Caution / JEE Pitfall:** Read questions carefully for keywords like 'incorrect', 'not true', or non-standard conditions.

■ **Speed Trick:** Elimination of extreme or unphysical answer choices improves accuracy in assertion-reason and multiple-statement questions.

Representative Source PYQs:

- [2023 Morning Shift] Q#1 (MCQ): "Which one of the following reactions does not occur during extraction of copper?..." [Ans: N/A]
- [2023 Morning Shift] Q#2 (MCQ): "Which one of the following is not an example of calcination?..." [Ans: B]
- [2023 Evening Shift] Q#2 (MCQ): "Which one of the following statements is incorrect?..." [Ans: N/A]

3.3 Isolation Of Elements — Standard Calculations & Problem Archetypes (Moderate Yield — 4 PYQs)

Core Principle: Quantitative numerical problem-solving techniques and standard question patterns frequently seen in Isolation Of Elements.

■ **Caution / JEE Pitfall:** Watch for algebraic rounding, stoichiometric balancing, and boundary condition checks.

■ **Speed Trick:** Dimensional consistency checks and ratio comparisons can quickly eliminate incorrect options.

Representative Source PYQs:

- [2023 Evening Shift] Q#2 (MCQ): "In Hall - Heroult process, the following is used for reducing Al_2O_3 :-..." [Ans: C]
- [2022 Morning Shift] Q#3 (MCQ): "Given below are two statements : Statement I : According to Ellingham diagram, any metal oxide with higher ΔG° is more sta..." [Ans: N/A]
- [2020 Shift] Q#4 (MCQ): "An Ellingham diagram provides information about :-..." [Ans: N/A]

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: JEE Main 2025 (Online) 28th January Evening Shift (Q#1)

The purification method based on the following physical transformation is : $\text{Solid} \xrightarrow{\leq} \text{Vapour} \xrightarrow{\leq} \text{Solid}$

- (A) Extraction
- (B) Crystallization
- (C) Distillation
- (D) Sublimation

Correct Answer: None

Solution Analysis: The purification method described by the transformation: $\text{Solid} \xrightarrow{\leq} \text{Vapour} \xrightarrow{\leq} \text{Solid}$ is known as Sublimation. In sublimation, a solid substance directly transforms into its gaseous phase upon heating and reverts back to the solid phase upon cooling. This technique is commonly used to purify substances, particularly those that sublime more easily than others under the same conditions. Sublimation is effective for separating substances with different

Benchmark Problem 2: JEE Main 2023 (Online) 25th January Morning Shift (Q#1)

Which one of the following reactions does not occur during extraction of copper?

- (A) $2\text{FeS} + 3\text{O}_2 \rightarrow 2\text{FeO} + 2\text{SO}_2$
- (B) $\text{FeO} + \text{SiO}_2 \rightarrow \text{FeSiO}_3$
- (C) $\text{CaO} + \text{SiO}_2 \rightarrow \text{CaSiO}_3$
- (D) $2\text{Cu}_2\text{S} + 3\text{O}_2 \rightarrow 2\text{Cu}_2\text{O} + 2\text{SO}_2$

Correct Answer: None

Solution Analysis: $\text{CuFeS}_2 + \text{O}_2 \rightarrow \text{Cu}_2\text{S} + \text{FeO} + \text{SO}_2$ + very small FeS + very small Cu_2O
 $\text{Cu}_2\text{S} + \text{O}_2 \rightarrow \text{Cu}_2\text{O} + \text{SO}_2$
 $\text{FeS} + \text{O}_2 \rightarrow \text{FeO} + \text{SO}_2$
 $\text{FeO} + \text{SiO}_2 \rightarrow \text{FeSiO}_3$
 No formation of calcium silicate (CaSiO_3) in extraction of Cu.

Benchmark Problem 3: JEE Main 2023 (Online) 8th April Evening Shift (Q#2)

In Hall - Heroult process, the following is used for reducing Al_2O_3 :-

- (A) Magnesium
- (B) CaF_2
- (C) Graphite **[Correct]**
- (D) Na_3AlF_6

Correct Answer: C

Solution Analysis: In the Hall-Héroult process, graphite anodes are used to reduce alumina (Al_2O_3). During the electrolytic reduction, Al^{3+} ions are reduced at the cathode to form aluminum, while at the anode, oxide ions are oxidized to form oxygen gas. The oxygen gas then reacts with the carbon in the graphite anodes to produce carbon dioxide. This means that graphite is consumed during the process, and the anodes have to be replaced periodically. So, in the context of the question, graphite (Option C) is the correct answer, as it is the material of the anode used for the electrolytic reduction of Al_2O_3 in the